Harshwardhan Pandey

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Professional Summary

Smart India Hackathon National Finalist and final-year CS engineer who has fine-tuned LLMs in production, built end-to-end data pipelines, and shipped a blockchain audit trail at Deloitte — all before graduating. Seeking a data engineering role where I can apply my background in ETL systems, SQL-backed data modeling, and AI to scalable, real-world infrastructure.

Education

B.Tech — Computer Science Engineering *2022 – 2026 (Expected)*
Manipal University Jaipur, Rajasthan

Technical Skills

Languages: Python, C++, SQL
Data Engineering: ETL pipelines, data profiling, fact and dimension tables, star schema / Kimball modeling, data warehousing
Databases and Visualization: PostgreSQL, MySQL, MongoDB, Power BI, Tableau, Excel
Cloud and Big Data: Google BigQuery (GCP), AWS (foundational), Snowflake (foundational)
AI and ML: LLM fine-tuning, prompt engineering, NLP, Explainable AI, Computer Vision, BERT, GANs
Workflow and Tools: Git, Agile, Jupyter Notebook, Google Colab, Kaggle, Magento

Key Achievements

- Smart India Hackathon — National Finalist, ranked among top teams from 1,000+ submissions nationwide.
- IEEE Hackathon — Finalist; recognised for innovative application of AI to real-world infrastructure challenges.
- Deloitte Capstone Program — Finalist; selected among top performers for a blockchain supply chain solution.

Work Experience

Full Stack and AI Developer — AgriCart Farmtech Private Limited *Jul 2025 – Jan 2026*

- Fine-tuned domain-specific LLMs on 8 GB of structured agricultural datasets, reducing chatbot hallucination rate by 35% for approximately 200 enterprise users and farmers across India.
- Built an end-to-end ETL pipeline that ingested, profiled, and validated raw agricultural data into model-ready training sets, cutting preprocessing time by 10 hours per week.
- Engineered a Magento-based B2B2C e-commerce platform serving 150 concurrent users, achieving 99.5% uptime through iterative performance monitoring and frontend optimisation.

Blockchain Developer — Deloitte (Capstone Program) *Apr 2025 – Jul 2025*

- Designed a blockchain-powered audit trail system using fact-and-dimension data modeling that reduced manual supply chain traceability queries by 40% across 6 nodes.
- Implemented smart contracts and an immutable ledger, cutting data reconciliation errors by 30% and improving auditability for 3 internal stakeholder teams.

Research Intern — IIT Roorkee (SPARK Program) *Jun 2023*

- Analysed 5 years of environmental datasets across 4 sustainability indicators using structured data profiling techniques, contributing findings to the SPARK programme research report.

Projects

Predicting Power Outages in Delhi — Weather and Substation Data

- Built an SQL-backed ETL pipeline joining 3 years of weather records with substation fault logs across a star schema; trained a gradient boosting model achieving 82% prediction accuracy for grid failure events.

Interpretable Disease Diagnosis

- Applied SHAP and LIME explainability techniques to a black-box ML classifier; performed data profiling on 12,000 patient records to identify and remove 18% data quality issues before model training.

Population Density Estimation — Satellite Imagery

- Designed an image ingestion and classification pipeline processing 2,000 satellite tiles; built a CNN model estimating population density across 5 categories at 78% accuracy.

Also built: *BERT-based sentiment classifier (91% accuracy, 50K reviews) | Vision Transformers vs. CNN benchmark across 4 datasets.*

Research

- Pneumonia Detection via Explainable AI — Grad-CAM visualisations on 5,800 chest X-rays to surface CNN decision boundaries for clinical review.
- GAN-LSTM Watermarking — Hybrid neural architecture achieving 12% SSIM improvement over CNN-only baseline in digital watermarking imperceptibility.
- DRL-Based UAV Navigation — Comparative survey of 25 papers on autonomous path planning, collision avoidance, and multi-agent coordination strategies.